# Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

## Average acceleration

| | |
|---|---|
| **Meaning in my words**<br>How would you explain average acceleration to a friend? | **How can you tell?**<br>Which two velocity readings would you subtract, and in what order? |
| **Example or sketch**<br>A cart changes from −2 to +4 m/s in 3 s. What is its average acceleration? | **Non-example and why**<br>A partner divides the whole distance by time. What quantity did that give? |

## Speeding up

| | |
|---|---|
| **Meaning in my words**<br>How would you explain speeding up to a friend? | **How can you tell?**<br>A cart changes from −2 to −5 m/s. What happens to its speed? |
| **Example or sketch**<br>Draw three equal-time dots for a cart speeding up to the right. | **Non-example and why**<br>A cart moves right with acceleration to the left. Is it speeding up before it turns? |

# Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

## Slowing down

| | |
|---|---|
| **Meaning in my words**<br>How would you explain slowing down to a friend? | **How can you tell?**<br>Velocity changes from −5 to −2 m/s. What happens to the speed? |
| **Example or sketch**<br>Draw velocity and acceleration arrows for a bike braking while moving right. | **Non-example and why**<br>“Negative acceleration always means slowing down.” What happens if velocity is negative too? |

## Reversal

| | |
|---|---|
| **Meaning in my words**<br>How would you explain reversal to a friend? | **How can you tell?**<br>On a velocity–time graph, how can you tell a cart actually reversed? |
| **Example or sketch**<br>A cart goes from +2 m/s through zero to −1 m/s. Draw its direction before and after. | **Non-example and why**<br>A bus stops, then keeps going east. Does stopping prove it reversed? |

# Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

## Draw the before and after velocity arrows (m/s). Mark the change in velocity (m/s). Divide by time to label the acceleration (m/s²).

Before

After

**My marks mean**

# Choose the speed alert

07

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

**Our goal:** Use a velocity change to find acceleration.

**CHOOSE ONE JOB TOGETHER**

**TELEMETRY ENGINEER** ☐

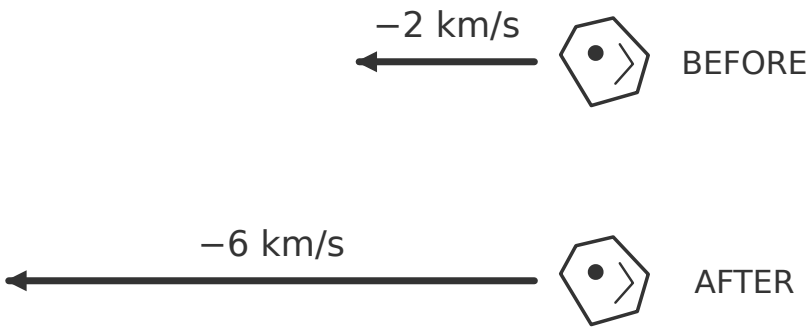
Radio the average acceleration with its sign and unit. Label both velocity arrows so the crew can check it.

**REPLAY COMMENTATOR** ☐

Narrate the two-second change. Point to the arrow lengths to explain whether XJ sped up or slowed down.

Point to a clue. Keep your first marks; add the update using the color key.

Two signed XJ velocities are **−2** and **−6 km/s**, two seconds apart. The control room must choose its speed alert.




## 01 OUR FIRST CALL

**Which speed alert should the control room use? Support it with the arrows.**

## 02 OUR REVISED CALL

**Wait for the reveal. Keep your first call.**

New clue: _____

Check the proposed banner. Give a corrected report using the two velocities.

**Switch speakers.** Why do two velocity readings give an average acceleration rather than the acceleration at every instant?

| ☐ GREEN | ☐ RED | ☐ YELLOW | ☐ BLUE | ☐ PURPLE | ☐ ORANGE | ☐ PENCIL |
|---|---|---|---|---|---|---|
| speeding up | slowing down | steady / balanced | position / displacement | velocity | forces | other marks |